IMPERIAL

A Modular Transfer Learning Pipeline for Medical Image Classification

Evaluated on breast cancer ultrasound (BUS-BRA) with CNNs, CLIP, DINOv2, and DINOv3

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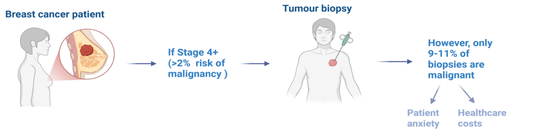
Introduction

Breast cancer is the most common cancer in women globally.[1,2] Early detection of breast cancer with ultrasound is critical as it can significantly improve clinical outcomes.[3] However, only 9–11% of biopsies performed following ultrasound screening are malignant, increasing healthcare expenses and patient anxiety.[4] Therefore, alternative non-invasive approaches for breast cancer classification are necessary.



Aims

- **Develop** a deep learning model pipeline to classify benign versus malignant breast ultrasound lesions
- **Compare** the performance of convolutional neural networks (CNNs) and vision transformer architectures for breast ultrasound classification
- **Evaluate** transfer learning strategies by comparing frozen versus fine-tuned backbones and the impact of segmentation-mask-guided training
- **Optimise** preprocessing methods, hyperparameters, and model architecture to maximize classification performance

Methods & Pipeline

We evaluated nine backbone architectures spanning **CNNs** and **vision transformers**, with frozen or fine-tuned weights, across three classification heads of increasing capacity. All configurations underwent hyperparameter search over learning rate and weight decay, with model selection by validation ROC-AUC and final reporting focused on malignant recall to minimize missed diagnoses.

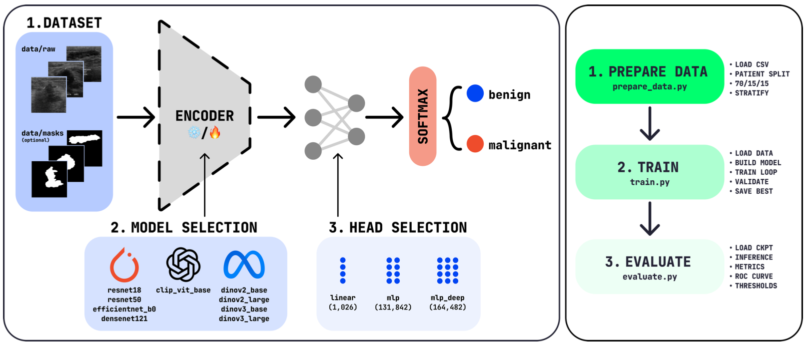
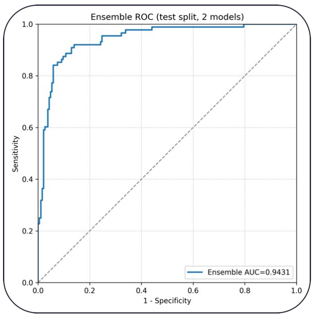


Fig. 1. (Left) Schematic Diagram of modular classification architecture. Input images (with optional lesion masks) are processed by an interchangeable encoder backbone (CNNs/ViTs) (frozen or fine-tuned) and heads enable systematic benchmarking.

Fig. 2. (Right) Three-stage training and evaluation pipeline. Patients are split 70/15/15 into training (1,316), validation (276), and test (274) sets with label stratification, maintaining an approximately 2:1 benign-to-malignant ratio. Models are trained with early stopping, saving the best checkpoint by validation AUC, then evaluated on the held-out test set with ROC analysis.

Results



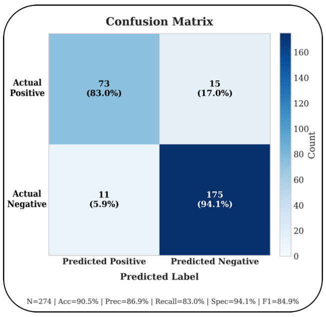


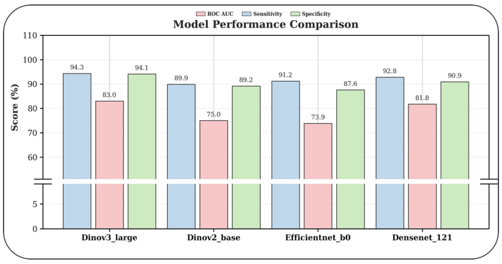
Fig. 3 & 4. Confusion matrix (left) and ROC curve (right) for the DINOv3-Large and DenseNet-121 ensemble (averaged outputs) on the held-out test set (AUC = 0.943). Both trained with lesion masks and unfrozen backbones.

- DINOv3-Large: lr=1e-5, wd=0.01, linear head
- DenseNet-121: lr=1e-4, wd=1e-4
- Both: bs=32

The curve rises steeply toward the upper-left corner, indicating strong discriminative ability across classification thresholds.

The best performing individual model, DINOv3-Large trained with lesion masks and an unfrozen backbone, achieved a ROC-AUC of 0.935. Assembling with DenseNet-121 (sensitivity 81.8%, the highest among individual models) via output averaging further improved performance, yielding a ROC-AUC of 0.943, accuracy of 90.9%, sensitivity of 84.1%, and specificity of 94.1%, with the confusion matrix and ROC curve shown below. The relatively high specificity reflects the dataset's 2:1 benign-to-malignant ratio; the operating threshold was selected to balance malignant recall against false alarm rate, in line with clinical screening priorities.

Fig. 5. Test set performance of the four best-performing individual models, reporting ROC-AUC, sensitivity, and specificity. DINOv3-Large achieves the highest AUC (0.935) and DenseNet-121 the highest individual sensitivity (81.8%), motivating their selection for assembling.



Clinical Validation

The **DINOv3 + DenseNet-121 ensemble** achieved an ROC-AUC of **0.943**, surpassing experienced (>18 years) and inexperienced (<3 years) radiologists. Whilst model sensitivity was marginally lower than clinicians, it significantly greater specificity enables high NPV predications, supporting DL as a decision-support tool for biopsy deferral in clinical obscure BI-RADS-4 tumours rather than a standalone replacement.

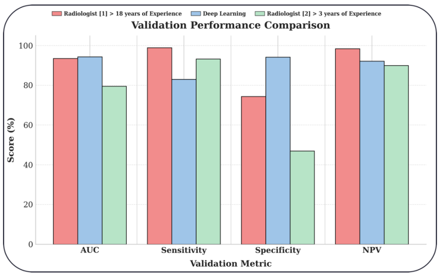


Fig. 6. Bar plot comparing validation metrics (AUC_ROC, Sensitivity, Specificity and Negative Prediction Value) of Dinov3_large model (blue bars) against radiologists with either less 3 (green bars) or greater than 18 years (red bars) of experience in the field[6]

Discussion & Limitations

Key Findings:

- Best single model (DINOv3-Large, fine-tuned, with masks) achieved ROC-AUC of 0.935
- Ensemble of DINOv3-Large + DenseNet-121 improved to ROC-AUC 0.943 (accuracy 90.9%, sensitivity 84.1%, specificity 94.1%)
- Complementary architectures (predicted): DINOv3 captures global context, DenseNet-121 extracts local texture patterns

Limitations:

- High specificity may reflect 2:1 benign-to-malignant class imbalance
- Threshold selection strategy optimised for balancing recall and false positives
- Single dataset evaluation (BUS-BRA) — external validation needed

Conclusion

- **DINOv3-Large + DenseNet-121 ensemble** achieved **ROC-AUC 0.943**, outperforming radiologists of all experience levels
- **Lesion masks** and **end-to-end fine tunings** were key to performance gains
- **Transformer-CNN ensembles** show strong potential as clinical decision-support tools for breast cancer screening

Future Directions

- **Optimise ensembles**, systematically explore architecture combinations to maximise complementary strengths
- Replace external lesion masks with **trainable segmentation modules** for fully integrated lesion detection and classification
- Evaluate on different, independent breast ultrasound datasets to **assess generalisability**
- Develop **interpretability tools** for radiologist decision support, integrate clinically

References:

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